

Dhruv Snehal Bhatt

Tempe, Arizona | (480) 919 6606 | dhruvsnehalbhatt.vercel.app | dbhatt24@asu.edu

EDUCATION: Arizona State University, Tempe, Arizona

B.S. in Computer Science (Dean's List)

Jan 2023 - May 2026

RELEVANT COURSES: Data Structures and Algorithms, Artificial Intelligence, Natural Language processing, Linear Algebra, IOS mobile App development, Software Engineering, Cyber Security, Applied statistics, Distributed software development.

SUMMARY

Computer Science senior who **builds and owns production systems end-to-end**, from **architecture and implementation to deployment and iteration**. As an **AWS Certified Cloud Practitioner**, I've delivered **AI-driven and cloud-backed applications used by real users**, with a focus on **reliability, scalability, and practical impact**. Comfortable working through ambiguity, designing services that hold up in production, and **improving systems based on real feedback**. I **move fast, learn continuously**, and take responsibility for outcomes, **not just code**, while building software meant to operate at **real-world scale**.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, C# (.NET)

AI / LLM Systems: LLM application development (OpenAI APIs), Prompt engineering, Retrieval-Augmented Generation (RAG), Semantic search, Embeddings, Hallucination control via context grounding

Backend & APIs: FastAPI, Flask, Node.js, RESTful API design, Webhooks (event-driven systems)

Data & Retrieval: FAISS (vector search), Document chunking & metadata pipelines, Weaviate (Vector Database)

Cloud & DevOps: AWS (Lambda, ECS Fargate, API Gateway, S3, DynamoDB, CloudWatch), Docker, Git, GitHub Actions, CI/CD, Kubernetes, Jenkins, Redis, Celery.

EXPERIENCES

Memorial Union, Arizona State University: Guest Services Associate Supervisor

June 2023 – August 2025

- **Lead a 16-person Guest Services team** across the Information Desk, Sparky's Den, and Lost & Found, coaching new staff and **aligning coverage with peak traffic**. **Managed refund processes** ensuring **customer satisfaction**
- **Built and deployed** a production-oriented **ASU Lost & Found web platform** to replace legacy campus workflows, through a **single role-based system** with clear operational separation between **UI and backend logic**.
- **Built an AI pipeline** that uses **Google Vision** to extract image labels and a **structured LLM** prompt flow to generate consistent titles and descriptions, **reducing manual entry** effort while improving listing quality and searchability.
- Designed security as a first-class constraint: **enforced server-side session validation + RBAC** (admin/staff), hardened the app against **OWASP Top 10 risks**, and validated fixes through **ZAP active/passive scans** with iterative remediation.
- Architected a cloud-ready **storage model using AWS S3** signed uploads with no PII in image metadata and only **hashed object references in the database**, locking down frontend with **strict CSP security** headers for production.

Expandus Consulting Pvt. Ltd: IT intern

Feb 2021 - Jan 2023

- Migrated large **Excel-based datasets into PostgreSQL** with a cleaner schema and indexed queries, cutting query response time by **30 percent** and improving **data accessibility** for reporting and decisions.
- **Designed and shipped RESTful APIs** to automate previously manual workflows, **reducing data entry effort by 50 percent** and improving **data retrieval speed by 40 percent** for internal users and clients.
- Partnered with cross-functional teams to deliver reliable **client integrations** and reduce handoff friction, improving system stability and **service delivery by 25 percent** through tighter validation and integration coordination.

PROJECTS

GitHub PR Review Assistant (TypeScript, Node.js, GitHub App, OpenAI) ([link](#))

- Built an **event-driven GitHub App** that analyzes pull request diffs on open/update events and generates high-signal review summaries and inline comments that **Cuts review time and reviewer fatigue**.
- Implemented **risk-focused analysis** to flag security-sensitive changes, complex logic, API contract shifts, and missing or insufficient tests, **enforcing consistent checks for risky logic and test gaps**.
- Designed reviews outputs to prioritize architecture, edge cases, and test coverage over boilerplate diffs, improving review consistency and **automated** posting proper feedback directly on PRs, **reducing manual review** overhead and triage time

ASU RAG-based LLM Information Chatbot (Python, FastAPI, FAISS, OpenAI) ([link](#))

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** system over ASU-specific documents to answer institutional queries with verified sources.
- Designed a **document ingestion pipeline** with chunking, metadata tagging, embedding generation, and **FAISS-based semantic retrieval** to return the most relevant passages for real user queries.
- **Reduced hallucinations** by generating response from retrieved context and returns no answer when coverage is missing.
- Architected the system for **rapid knowledge updates** (add docs → re-index → improved answers) without **model retraining**, making it easy to change and update the files without fine tuning the model repeatedly.

CERTIFICATIONS AND AWARDS

- AWS Cloud Practitioner
- CFI Financial Modelling and Valuation Analyst
- NaMU Scholarship of \$58,000
- Dean's List of High Achievers